

Slab generation: why FCC metals hit walltime / OOM, and the fix

codes/files/slab_generate.py (SurfaceBuilderWorkChain)

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1 Symptom

The SlabGen CalcJob of SurfaceBuilderWorkChain (one job covers all ~10 bulks of a formula) had an **undetermined walltime** and, for plain FCC metals such as **Ag** and **Cu**, was killed either at the 6 h wall or by an **OOM** event — even though those elements have a 1-atom primitive cell and should slab in milliseconds.

2 Diagnosis

It is not the element, it is the *symmetry of the structure that reaches the runner*. The bulks fed to slab generation come out of the generation / CSP / MinimaHopping stage already **MLIP-relaxed**, so they carry ~0.001–0.01 Å of numerical noise (and the generator deliberately explores distorted polymorphs).

generate_for_structure standardized with the tight default symprec=0.01 (SpacegroupAnalyzer(structure, symprec=0.01, ...)). At that tolerance a *perfect* FCC metal with a hair of MLIP noise is mis-read as a large **P1** cell. The consequences inside generate_all_slabs:

- in P1 **every atomic layer is a distinct termination**, so it enumerates dozens of Miller planes × many spurious shifts;
- with max_normal_search = max_miller_idx it builds a **large oriented supercell** for each plane to orthogonalise the c-axis;
- all those slabs, for **all bulks**, are held in RAM in a **single** CalcJob that shares its memory budget with the adsorbate generator/relaxer.

That is the walltime blow-up *and* the OOM.

2.1 Benchmark (Ag, max_index = 2)

Input to the runner	spglib sees	generate_all_slabs
Clean FCC Ag (Fm-3m, 4-atom conv.)	Fm-3m, 4 at	6 slabs, 3.2 s, 103 MB
FCC Ag as 2×2×2 + 0.01 Å noise, at symprec=0.01	P1, 32 at	(pathological)
Same cell, standardized at symprec=0.05	Fm-3m, 4 at	6 slabs, 3.2 s
Genuinely distorted P1, 32 at	P1, 32 at	TIMEOUT > 180 s

A clean supercell is *not* the problem — spglib reduces it back. Noise is.

3 The fix

All in codes/files/slab_generate.py; the slab metadata contract is unchanged (still a pymatgen Slab → pmg_to_ase, carrying miller_index / shift / scale_factor / oriented_unit_cell to slab_relax.py).

1. **Symprec-ladder standardization** (_standardize_bulk). Walk SYMPREC_LADDER = (0.1, 0.05, 0.01) (loose → tight) and keep the **smallest** conventional cell **whose reduced composition is preserved**, so a noisy metal collapses back to its 4-atom Fm-3m cell while an alloy is never over-merged into a different stoichiometry. Falls back to the original tight symprec=0.01 cell if no rung preserves the composition.
2. **max_normal_search = 1** (was = max_miller_idx). Orthogonality is already enforced afterwards by process_slab (it drops any slab with α or β more than 1° off 90°), so a costly normal search is wasted work.
3. **Guards** so one bad cell cannot take down the shared multi-bulk job:
 - **size + symmetry skip** — skip generation only when the standardized cell is **both** large (> MAX_CONV_ATOMS = 60 atoms) **and** low-symmetry (spacegroup number ≤ LOWSYM_SG_MAX = 15, i.e. triclinic/monoclinic). That is exactly the OOM/hang regime and is not a meaningful slab target.
 - **per-bulk walltime cap** — generate_all_slabs runs under a SIGALRM_time_limit(GEN_TIMEOUT_S = 600 s); a slower cell raises SlabGenTimeout, which run_slab_generate already catches per structure and turns into an empty result (skip), not a job-wide failure.

3.1 Legitimate oxides are not affected

The skip needs *low symmetry* as well as size, so an ordered oxide stays in. A Fd-3m pyrochlore has an 88-atom conventional cell but spacegroup #227 >> 15, so it passes; and because it is high-symmetry it has few distinct Miller planes and terminations, so generate_all_slabs is well behaved on it anyway.

3.2 Verification

/tmp/test_slabgen.py style checks (all pass):

- noisy 32-atom Ag → standardized to 4-atom Fm-3m → 6 slabs in ~2.5 s;
- _time_limit raises a catchable SlabGenTimeout;
- an AgCu alloy keeps its reduced formula (no over-merge);
- a distorted 108-atom P1 cell (sg #1) is skipped by the guard.

4 Knobs (in slab_generate.py)

Constant	Default	Meaning
SYMPREC_LADDER	(0.1, 0.05, 0.01)	loose→tight symprec rungs for standardization
MAX_CONV_ATOMS	60	size half of the skip test
LOWSYM_SG_MAX	15	spacegroup-number half of the skip test
GEN_TIMEOUT_S	600	per-bulk walltime cap on generate_all_slabs

5 Not done here (deliberately)

A lighter, symmetry-free testing backend (ASE ase.build.surface: ~ms per facet, flat memory, immune to input symmetry, but one termination per facet and no oriented-cell metadata) was considered and **not** added — this fix keeps the production pymatgen path and its full termination enumeration.